

CubeSat Electrical Power System (EPS)

Standard Operating Procedure (SOP)

Mission EPS Group

2026-05-11

Document Control

Role	Name	Date
Prepared By	Mission Engineering Team	2026-05-11
Reviewed By	EPS Lead Engineer	2026-05-11
Approved By	Flight Director	2026-05-11

Document ID: CS-EPS-SOP-001

Version: 1.0

Status: Formal Mission Document

Date: 2026-05-11

1 Introduction

This document defines the standard operating procedures for the CubeSat Electrical Power System (EPS). It covers startup, health monitoring, nominal operations, and emergency response. All operators must adhere strictly to these steps to ensure mission longevity.

2 Satellite Startup Procedure (Initialization)

Objective: Safely transition the satellite from cold-launch state to Nominal Operations.

Step	Action	Conditions	Expected Output	Verification
1.1	BATT_ACTIVATE	Deployment switches closed	Battery voltage > 10.0V	EPS Telemetry <code>Input_Voltage</code> active
1.2	INITIAL_BOOT	Solar panels exposed	Input Voltage stabilizing	<code>simulated_eps.input_voltage</code> > 10.0V
1.3	RAIL_3V3_ENABLE	Startup delay +30s	3.3V rail voltage 3.3V	Rail Status: 3.3V: OK
1.4	MODE_TRANSITION	SoC > 20%	Mode: SAFE	<code>eps_controller.mode</code> == "Safe"
1.5	RAIL_5V_ENABLE	SoC > 40%	5.0V rail voltage 5.0V	Rail Status: 5V: OK
1.6	NOMINAL_ENTRY	All vital rails active	Mode: NOMINAL	<code>eps_controller.mode</code> == "Nominal"

3 Power System Health Check Procedure

Objective: Periodic verification of EPS integrity and component health.

Step	Parameter	Nominal Range	Verification Tool
2.1	Bus Voltage	10.0V – 14.4V	<code>get_status()["input_voltage"]</code>
2.2	3.3V / 5V / 12V	±5% Nominal	<code>get_status()["rails"]</code>
2.3	Rail Temperature	< 60°C	<code>rail.temperature</code>
2.4	Total Current	< 5.0A (Aggregate)	<code>sum(rail.output_current)</code>
2.5	State of Charge	40% – 95%	<code>battery.soc</code>

4 Data Downlink Procedure (COMMS Mode)

Objective: Manage power resources during high-drain communication windows.

Step	Action	Conditions	Expected Output	Verification
3.1	PRE_DOWNLINK_CHECK	Ground station in view	SoC > 50%	Proceed to Step 3.2
3.2	PAYLOAD_SHED	Preparing for Comms	Non-essential rails OFF	12V Rail: Disabled
3.3	COMMS_ACTIVE	Mode: COMMS	Radio Power > 5.0W	High 5V current draw observed
3.4	REALTIME_MONITOR	During downlink	SoC > 25%	No Undervoltage faults
3.5	POST_COMMS_RESET	Downlink complete	Return to NOMINAL	Mode: NOMINAL

5 Anomaly Response Procedure (ARP)

Objective: Mitigate system damage during hardware or environmental faults.

5.1 Thermal Shutdown

- **Condition:** Rail Temperature $> 80.0^{\circ}\text{C}$.
- **System Action:** EPS Controller disables the affected rail.
- **Procedure:**
 1. Identify affected rail via `fault_history`.
 2. Disable all non-essential loads.
 3. Wait for temperature to drop below 50.0°C .
 4. **Verification:** Check `rail.fault == None`.

5.2 Overcurrent Fault

- **Condition:** `requested_current > max_current`.
- **Procedure:**
 1. Immediately switch to **SAFE** mode.
 2. Check `simulated_load.py` for component failure simulation.
 3. Sequentially re-enable rails to isolate the short circuit.

5.3 Critical Low Power

- **Condition:** SoC $< 20\%$.
- **Procedure:**
 1. **SAFE_MODE_FORCED:** System automatically cuts 5V and 12V rails.
 2. Orient solar panels to Max Power Point (MPP).
 3. Disable all beaconing except *Distress Signal*.
 4. **Recovery:** SoC must reach 40% before NOMINAL transition.

6 Approval and Sign-off

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